Lab 1 – Requirements Table

Problem Statement #46 — Remote Team Time-Tracking & Project Approver
Student: M. Louis Alan | Actors: Remote Developer, Engineering Manager

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow developers to log time against specific project tasks and prevent timesheet submissions exceeding 24 hours in a single calendar day.	High	Pass: Valid timesheet submitted for manager approval. Fail: Timesheet logging 30 hours in one day is accepted.	Prevents fraudulent or erroneous time entries and protects payroll and budget accuracy. (Given)
FR-002	Functional	The system shall allow developers to associate each logged time entry with a valid Jira ticket ID by syncing with the connected Jira project management system.	High	Pass: A time entry linked to an existing, open Jira ticket ID is accepted and displays the ticket title. Fail: A time entry referencing a non-existent or closed Jira ticket ID is accepted without an error.	Ensures traceability between logged developer effort and actual project work items, enabling accurate budget attribution.
FR-003	Functional	The system shall allow developers to submit a completed weekly timesheet to their assigned Engineering Manager for approval.	High	Pass: A submitted timesheet appears in the manager's pending-approval queue within 5 seconds. Fail: A timesheet with missing daily entries is submitted without a warning.	Formalizes the approval workflow required before hours can be counted toward payroll and project accounting.
FR-004	Functional	The system shall allow Engineering Managers to approve or reject a submitted timesheet, requiring a mandatory comment when a timesheet is rejected.	High	Pass: A rejection attempted without a comment is blocked until the manager enters one. Fail: A rejection is accepted with an empty comment field.	Ensures developers receive actionable feedback when a timesheet is not approved, supporting quick correction and resubmission.
FR-005	Functional	The system shall display, for each active project, a real-time budget burn-rate chart comparing costed logged hours against the approved project budget.	Medium	Pass: The burn-rate chart reflects a newly approved timesheet's hours within 500 ms of approval. Fail: The chart shows stale data more than 500 ms after approval or omits an approved entry.	Gives Engineering Managers visibility into project spend so budget overruns can be caught early.
NFR-001	Nonfunctional (Performance & Security)	Timesheet approval status transitions and project budget burn-rate charts must update across all connected dashboards within 500 ms.	High	Pass: Benchmarking tests confirm the 500 ms latency target and required security standards are met under simulated peak load.	Real-time dashboards are required so managers can make budget decisions promptly. (Given)
NFR-	Nonfunctional	The system shall restrict timesheet-approval actions	High	Pass: A user with the Remote Developer	Protects salary-

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002	(Security / Access Control)	and project budget-burn views to users authenticated with the Engineering Manager role, enforced via role-based access control (RBAC).		role attempting to open the approval queue or budget dashboard receives an Access Denied response. Fail: A Remote Developer account can view or approve another developer's timesheet.	linked and budget-sensitive data from unauthorized access.

Note: FR-001 and NFR-001 were provided in the problem statement; FR-002–FR-005 and NFR-002 were authored to satisfy the deliverable requirement of exactly 5 FRs and 2 NFRs.

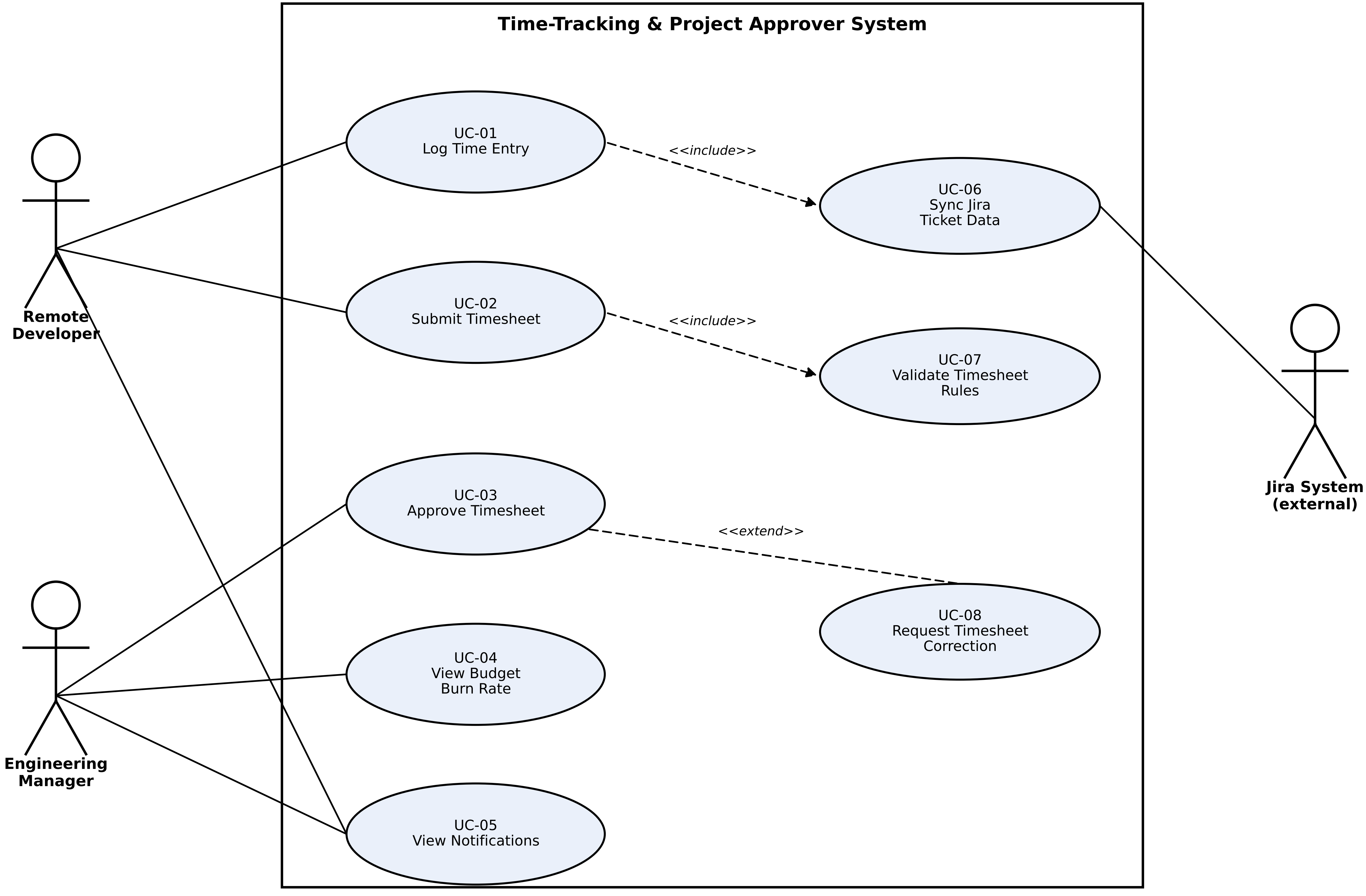

Fig. 1 — UML Use-Case Diagram: Remote Team Time-Tracking & Project Approver (Problem Statement #46)

Use-Case Flow Specification

Problem Statement #46 — Remote Team Time-Tracking & Project Approver

Student: M. Louis Alan

Use Case: UC-03 Approve Timesheet

Primary Actor: Engineering Manager

Secondary Actor: Remote Developer (notified of outcome)

Trigger: A Remote Developer submits a completed weekly timesheet (UC-02)

Preconditions:

- Remote Developer has submitted a complete weekly timesheet; its status is "Pending Approval".
- Engineering Manager is authenticated and holds the "Engineering Manager" role (NFR-002).
- The submitted timesheet passed the daily 24-hour cap validation at submission time (FR-001).

Postconditions:

- Success: Timesheet status = "Approved"; hours are reflected in the project budget burn-rate dashboard within 500 ms (NFR-001); Developer is notified.
- Failure: Timesheet status = "Rejected"; Developer is notified with the manager's mandatory comment (FR-004); Developer may revise and resubmit.

Main Success Scenario:

1. Engineering Manager opens the pending-approval queue.
2. System displays submitted timesheets with developer name, project, total hours, and linked Jira ticket IDs.
3. Manager selects a timesheet to review.
4. System displays timesheet details, including daily hours, task descriptions, and Jira ticket links (from UC-01/UC-06).
5. Manager reviews the entries and selects "Approve".
6. System re-validates the timesheet against the daily 24-hour cap and confirms the manager's RBAC permission.
7. System updates the timesheet status to "Approved".
8. System updates the project budget burn-rate dashboard within 500 ms (NFR-001).
9. System notifies the Remote Developer that the timesheet was approved.
10. Use case ends successfully.

Alternate Flow — 5a. Timesheet Rejected:

a) At step 5, Manager selects "Reject" instead of "Approve".
b) System prompts the manager for a mandatory rejection comment (FR-004) — this is the extension point for UC-08 "Request Timesheet Correction".
c) Manager enters a comment and confirms the rejection.
d) System validates that the comment field is not empty; if empty, the system blocks the action and re-prompts the manager.
e) System sets the timesheet status to "Rejected" and stores the manager's comment.
f) System notifies the Remote Developer with the rejection reason.
g) Developer revises the timesheet and resubmits it, returning to UC-02 "Submit Timesheet".
h) Use case ends.